Groundwater Buffering of Vegetation Function Through Drought in the High Plains (Ogallala) Aquifer

Thresholds, reversibility, mechanism and sustainability: a three-paper doctoral research plan (revision 2)

20 October 2026 · Revises *Groundwater Buffering of Ecosystem Function During Drought* (original plan) · Working repository *G:\MSU_GWB\datasets*

Summary. Vegetation over the High Plains aquifer can keep transpiring, photosynthesising and holding canopy water through meteorological drought where it reaches water that rain did not supply: a shallow water table, or water pumped from the aquifer. This plan asks how large that buffering is, where it fails, whether failure reverses when water returns, how the water, carbon and energy signals of vegetation come apart near failure, and whether the buffering is sustained by renewable water or by mining a declining aquifer (Scanlon et al. 2012; Haacker et al. 2016; Butler et al. 2016). It keeps the five research questions of the original plan and reorganises the work into a foundation stage and three papers. Two changes in approach drive the revision. First, managed and natural water use are separated by a *dynamic*, monthly, per-pixel state that is read from how vegetation and evapotranspiration (ET) depart from what rain alone would support, instead of an annual or static irrigation mask (Paper 1). Second, the expected-function model behind the buffering residual is rebuilt: it is trained only on vegetation that can draw on nothing but rain, predicts ET, greenness and canopy water jointly with calibrated intervals, and is checked for transfer to every pixel it is applied to; previously trained models are not used. Paper 2 estimates the buffering effect of groundwater access with designs that stay inside one climate band, which the existing shallow and deep groundwater clusters show to be necessary (their aridity index is 0.52 against 0.30), and classifies buffered systems as sustainable, mined or irrigation-subsidised using volumes. Paper 3 combines the threshold, reversibility and mechanism questions.

How to read this plan

- Every step has an identifier: **S0.x** for the shared foundation, **P1.x**, **P2.x**, **P3.x** for the papers. Each step card states its goal, inputs, method, output, a **verification test with a pass criterion** (shaded row), and what it depends on.
- Hypotheses (H1.1 …) and the numeric pass criteria are **proposals** to be fixed in the registered analysis plan (S0.7) before test data are examined. Numbers are taken from the literature only where a citation is attached.
- Decision gates G0–G3 (Section 9) state what changes if a verification fails.
- All 144 cited works were checked on 2026-10-20: 143 against Crossref by DOI (journal version; title, first author and year agree) and one conference paper in OpenAlex. Abstracts were read for the papers that carry a specific claim here; the reader should still confirm any number before quoting it.

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1. Research questions and what changed

The five research questions of the original plan are unchanged. Table 1 maps them to papers and steps; Paper 1 is the enabling paper that produces the signal the other two use.

Research question	Paper	Steps
RQ1 Buffering. Does groundwater access measurably sustain dry-season ET, photosynthesis and canopy water during drought, relative to comparable vegetation without access?	2	P2.1–P2.6
RQ2 Threshold. At what depth to water, rate of decline or storage anomaly does buffering sharply weaken?	3 (Part A)	P3.1–P3.2
RQ3 Reversibility. When water returns, does function recover, or does it show hysteresis?	3 (Part B)	P3.3–P3.5

Research question	Paper	Steps
RQ4 Mechanism. Near the threshold, do water, carbon and energy signals decouple, and does the order depend on plant functional type?	3 (Part C)	P3.6–P3.8
RQ5 Sustainability. Is buffering sustainable use of stored water, groundwater mining, or irrigation subsidy?	2	P2.7
Enabling: when and where does vegetation use water that rain did not supply, and from which source?	1	P1.1–P1.7

Table 1. Research questions, papers and steps.

Topic	Original plan	Revision 2	Why
Study area	High Plains first, then Central Valley, Michigan / Corn Belt, Mediterranean and Murray–Darling basins	High Plains (Ogallala) only; other basins move to future work (Section 12)	Your decision; one aquifer with dense well, crop and management records
Analysis scale	Harmonised ~1–5 km grid; decoupling at 0.1–0.25°	Two scales: 1/24° lab gridMET cells aquifer-wide and 480 m blocks of the 30 m Landsat / OpenET lattice; native 30 m in focus landscapes	30 m indices and OpenET are now available; 480 m = 16 × 16 pixels nests exactly, so no resampling
Managed vs natural water	Stratify with static maps (CDL, LANID, GFSAD)	Monthly per-pixel subsidy state read from vegetation and ET departures; static maps and pumping records only validate it	Fields switch between irrigated and dryland across years (Deines et al. 2017); a static mask is least reliable in the drought months that matter
Expected function	ET from climate and soil moisture, one model fitted on all pixels	Rain-only reference model: trained on vegetation that can use only rain; predicts ET, greenness and canopy water with calibrated intervals; transfer checked; two independent estimators for robustness	A model fitted on all pixels learns subsidised behaviour through proxies and absorbs the signal; the audited models are not used
Causal design	Matching across the aquifer	Contrasts inside climate strata: matching within aridity bands, valley–upland transects, alluvial-boundary discontinuity, within-unit difference-in-differences, the Sheridan-6 LEMA	The shallow and deep clusters differ in climate as much as in depth (Table 4)
Work plan	Phase 0 pilot, Phases 1–2, advanced extension	Step 0 and three papers; thresholds, reversibility and mechanism combined in Paper 3	Your decision
Specific yield	Uniform S = 0.15 in the TBI work	Gridded specific yield (to acquire), 0.10–0.20 as sensitivity	Volume checks scale linearly with specific yield

Table 2. Changes from the original plan.

2. Starting point: what already exists

2.1 Data holdings

The plan uses the holdings below (checked in the repository and its 20 October 2026 data audit). Periods are those of the files or notebooks; where an export is unfinished it is noted.

Signal	Product (repository)	Grid · period	Role in this plan
Vegetation state	Landsat Collection 2 Level-2, 18 indices, monthly median composites (30 m notebook)	30 m UTM-zone tiles · 2005–	Greenness, canopy water, open water; core function signal
ET (field scale)	OpenET v2.1 ensemble and six models (Melton et al. 2022; Volk et al. 2024)	30 m · 1999-10 to 2026-09 monthly	Core ET signal; inter-model spread as uncertainty

Signal	Product (repository)	Grid · period	Role in this plan
ET (coarse)	SSEBop MODIS dekadal (Senay et al. 2013)	1 km · 2003 to 2022-05	Cross-check, longer coarse ET
ET, canopy temperature	ECOSTRESS L3T JET v002, ECHO-ET (Fisher et al. 2020)	70 m · 2018-07–	Dry-down detail; canopy temperature (merged 0.01° file covers 2018-07 to 2020-08 only)
Photosynthesis	OCO-2 and OCO-3 Lite SIF soundings (Sun et al. 2017); GOSIF (Li & Xiao 2019)	Soundings; 0.05° · 2014–; 2000–2025	Independent carbon signal (OCO); GOSIF is derived from MODIS and meteorology, so context only
Canopy water (microwave)	SMAP L3 enhanced VOD, AM and PM	9 km · 2015-03–	Water leg of decoupling; plant hydraulic strategy
Root-zone soil moisture	SMAP L4 (Reichle et al. 2019)	9 km, 3-hourly · 2015-03 to 2024-05	Supply driver and check of the soil bucket (S0.3)
Long greenness	MODIS MOD13A3	1 km monthly · 2000–	Long record for reversibility
Meteorology, drought	gridMET daily and drought indices, Climatology Lab files (Abatzoglou 2013)	1/24° · 1979– (daily), 1980– (pentads)	Drivers; event catalogue
Storage	GRACE / GRACE-FO mascons (Watkins et al. 2015)	~0.5° · 2002-04 to 2024-09	Basin-scale storage constraint
Groundwater	HPA_wells_water_level_timeseries.nc (USGS, NECSD, TWDB); USGS campaign wells; HRES-WTD (Ma et al. 2026); USGS water-level change maps	Points · 2000–2026; 30 m static (2015)	Exposure, dynamics, rebound selection
Soils	gSSURGO stack (rootznaws, aws, soc, nccpi)	120 m · static	Available water capacity, productivity
Crop type	USDA NASS CDL (Boryan et al. 2011)	30 m · 2008–2025	Vegetation class for the reference model; validation, never a mask
Terrain	GLO-30 DEM	30 m · static	HAND and topographic position (mask the ~370 km² zero block in eastern Nebraska)
Groundwater-dependent ecosystems	GDE map (Rohde et al. 2024b)	30″ · static	Labels for natural subsidy
Aquifer properties	Hydraulic conductivity, PAW, water-level change (derived 4 km)	Old 287 × 181 grid · static	Covariates after remapping to the lab grid
Flux towers	AmeriFlux records	Points · 2005–2021	ET and GPP validation
Woody structure	GEDI L4B biomass	1 km · 2019-04 to 2023-03	Woody fraction covariate; VOD interpretation

Table 3. Data holdings used by the plan.

2.2 Prior analyses and their status

- **Static vulnerability index (TBI v3).** The VVIP index of Gonzalez Cruz et al. (2021) was rebuilt with real gSSURGO plant-available water and USGS hydraulic conductivity; two terms remain placeholders. It is used here only as a descriptive covariate map.
- **Residual-ET models (XGBoost, LSTM-HBV).** Audited on 2026-10-07 (29 defects; reported numbers withdrawn). As you asked, they are not used. Their leakage-safe design rules are kept: strict time lookups, per-cell baselines ending before the model period, blocked folds with an embargo, an in-house standardised precipitation index, and the tower screening.

- **Shallow and deep groundwater clusters.** Local Moran clusters (Anselin 1995) of 2019 well depth to water on 0.25° × 0.30° cells (Figure 1), plus a yearly 2000–2017 persistence version. The shallow cluster covers 9.9 % of the aquifer, the deep cluster 20.0 %; cells in the same cluster in at least 80 % of years cover 8.6 % and 17.0 %.

2.3 What the shallow and deep map implies

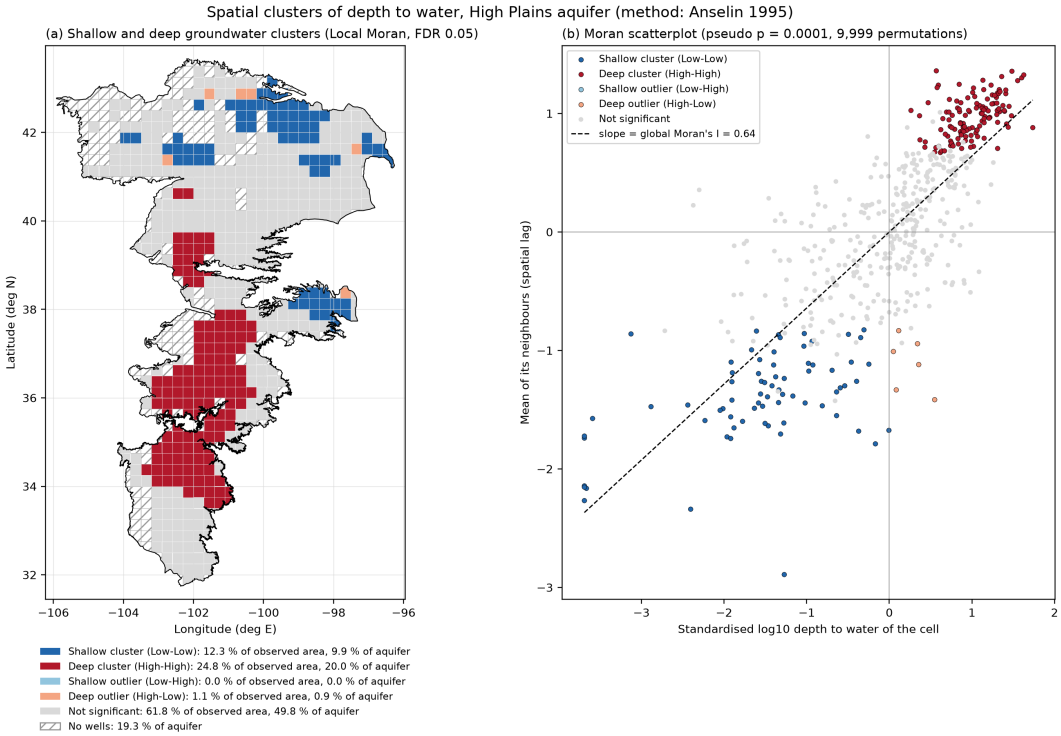


Figure 1. Existing shallow (blue) and deep (red) groundwater clusters, 2019 wells, Local Moran with false-discovery-rate control (repository: plots/gw_shallow_deep_clusters/fig2_shallow_deep_clusters.png).

Table 4 compares the two clusters. They differ in every climate variable with large effect sizes, and in decline since predevelopment. A direct shallow-versus-deep comparison therefore measures groundwater access *plus* climate *plus* depletion history. Because the clusters barely overlap in climate, matching across them cannot remove the difference; estimates would rest on extrapolation outside the common support (Crump et al. 2009).

Variable (cluster median)	Shallow	Deep	Rank-biserial r
Depth to water, 2019 (m)	4.0	67.8	1.00
Precipitation (mm per year, 1981–2010)	643	483	−0.73
Reference ET (mm per year)	1,230	1,648	0.97
Aridity index P / PET	0.52	0.30	−0.96
Vapour pressure deficit (kPa)	0.77	1.12	0.98
Mean temperature (°C)	9.3	13.6	0.83
Water-level change, predevelopment to 2019 (m)	+0.16	−13.3	−0.98
Water-level change, 2017 to 2019 (m)	+0.15	−0.10	−0.87
Plant-available water (mm)	136	232	0.63
Hydraulic conductivity (m per day)	11.2	22.2	0.48
Cells	72	127	

Table 4. Shallow versus deep cluster (repository: table2_shallow_vs_deep_tests.csv; Mann–Whitney $p < 10^{-7}$ in every row).

The map is therefore used in three ways: (i) as a stratum within which every result is reported; (ii) to locate landscapes where shallow and deep water tables occur *inside one climate band*, chiefly river valleys cut into uplands (Section 3.3); (iii) through its persistence version, to separate stable from shifting depth classes in Paper 3.

3. Cross-cutting design

3.1 Rules that apply to every step

- **R1 Function first.** Buffering is measured on vegetation function: ET, greenness, canopy water, photosynthesis and canopy temperature. Hydrology (wells, depth to water, storage) enters as exposure, stratum or volume check, never as the outcome.
- **R2 Exposure never enters the expected-function model.** Depth to water, height above nearest drainage (HAND), well trends, irrigation labels, distance to streams and the GDE map are excluded from its inputs; an automated test refuses them.
- **R3 Out-of-sample everywhere.** Every residual is out-of-fold. Folds are spatial blocks crossed with time blocks with an embargo (Roberts et al. 2017; Ploton et al. 2020).
- **R4 Two scales, one definition** (Section 3.2). Headline results are reported at both scales; disagreement is a finding, not noise (Jelinski & Wu 1996).
- **R5 Contrasts inside climate strata.** Exposed and unexposed vegetation are compared only within climate strata; overlap is diagnosed and units outside the common support are trimmed (Crump et al. 2009).
- **R6 Registered tests.** Hypotheses, outcomes, pass criteria and robustness checks are written down (S0.7) before test folds are examined; families of tests across strata use false-discovery-rate control (Benjamini & Hochberg 1995).
- **R7 Traceable numbers.** Every figure and table comes from a script and a manifest; intervals come from a block bootstrap over space and time.
- **R8 Data as stored.** Source files are never modified; quality flags are applied in the analysis layer and their effect is reported.

3.2 Two scales

Scale	Grid	Aggregates carried	Used for
C coarse	Lab gridMET 1/24° (287 × 233); 27,333 aquifer cells with ≥ 50 % inside hp_bound2010	Cell mean, within-cell SD, valid count; state and class fractions from 30 m	Aquifer-wide inference; linking SIF, VOD and GRACE; Paper 3 mechanism
F fine	480 m blocks (16 × 16 pixels) on each UTM zone's 30 m lattice (zones 13, 14, a little of 15); about 2.0 million blocks for ~455,000 km²	Mean, SD, valid count; fraction of pixels in each monthly state	Paper 1 states aquifer-wide; Paper 2 designs; Paper 3 thresholds
N native	30 m Landsat / OpenET / CDL pixels; ECOSTRESS 70 m	—	Focus landscapes: field edges, riparian transects, attribution

Blocks that straddle a zone boundary exist in both zones and are flagged; no pixel is resampled. Headline estimates are repeated on 960 m blocks (32 × 32 pixels) to show how the answer depends on scale. Coarsening to 480 m reduces pixel-level noise (cloud residue, mixed field edges) while keeping single centre-pivot fields resolvable.

3.3 Focus landscapes

Seven landscapes are processed at native resolution first. They are chosen so that each paper has contrasts inside one climate band; you may replace or add sites (Section 13).

ID	Landscape	Groundwater and land use	Contrast it offers
FL1	Central Platte River valley and Sandhills margin (NE)	Shallow cluster; wet meadows, riparian woodland, irrigated valley	Valley versus upland in one climate; natural versus irrigated subsidy
FL2	Nebraska Sandhills interior	Shallow cluster; grassland, lakes and wet meadows; little irrigation	Natural subsidy with minimal management
FL3	Republican River basin (NE, KS, CO)	Irrigated, rain-fed and riparian land along one river; gradient of well decline	Decline gradient; stream–aquifer links
FL4	Arkansas River corridor and Great Bend Prairie (KS)	Shallow cluster near 38° N; riparian corridor	Riparian function versus river and well levels
FL5	Sheridan-6 LEMA and surroundings (northwest KS)	Pumping restriction since 2013 (Deines et al. 2019b; Drysdale & Hendricks 2018)	Management natural experiment
FL6	Southern High Plains playa country (TX)	Deep cluster; playas focus recharge (Gurdak & Roe 2010); dryland and irrigated cotton	Irrigation subsidy under heavy depletion; playa hydroperiod
FL7	Canadian River breaks (TX, OK)	Riparian corridor cut through deep-water-table uplands	Valley versus upland inside the hot, deep climate band

3.4 Periods

Monthly is the common step (Landsat composites and OpenET are monthly); drivers are aggregated from daily gridMET. **Core period 2008–2025**, where CDL, Landsat 30 m and OpenET overlap. **Long period 2000–2025** for reversibility (MODIS greenness, OpenET from 1999-10, GOSIF to 2025, wells). **Multi-sensor period 2015–2025** for mechanism (SMAP from 2015-03, OCO-2 from 2014-09, OCO-3 from 2019-08, ECOSTRESS from 2018-07). The 2012 and 2022 droughts (Hoerling et al. 2014) and the 2018–2019 wet years are the anchor events.

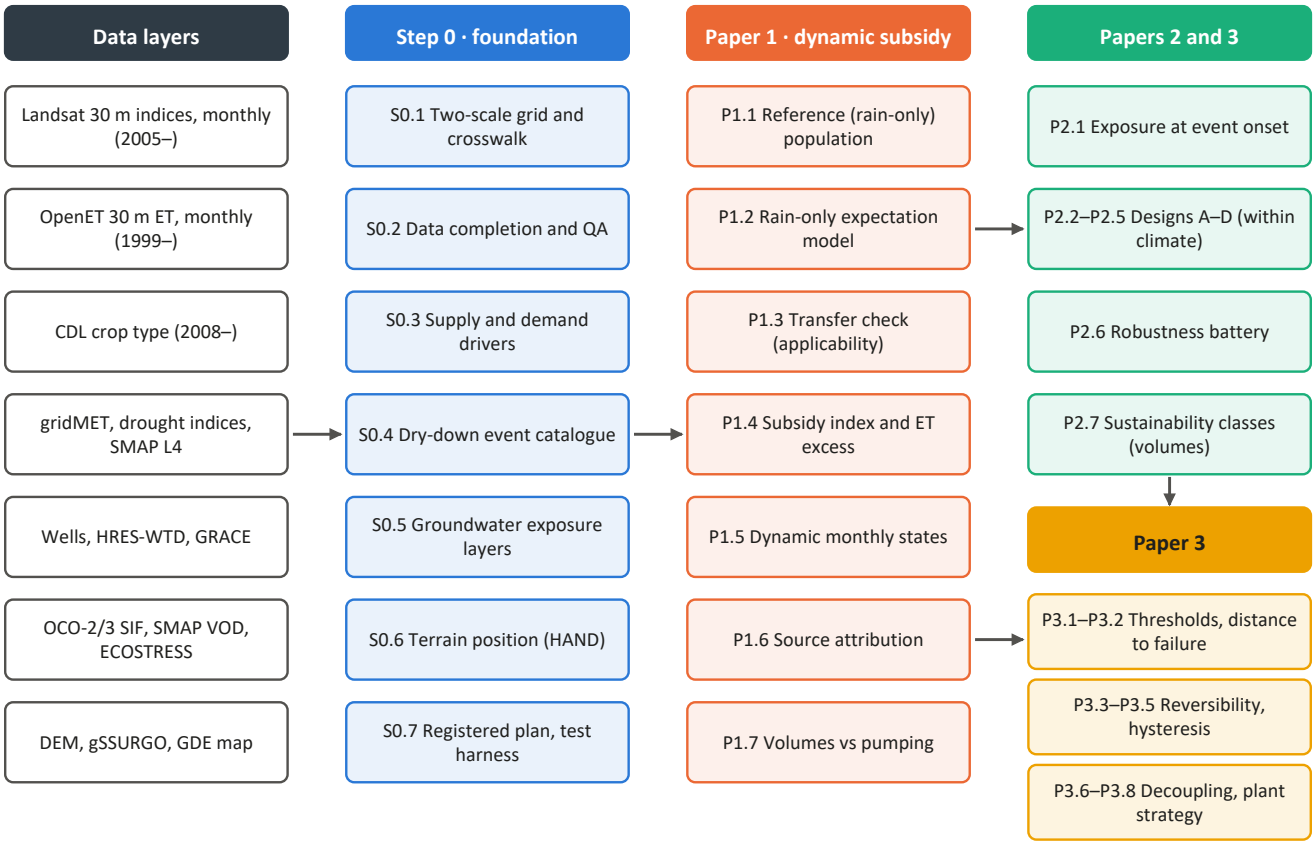


Figure 2. Workflow. Arrows show which outputs feed which steps; Paper 3 also uses Paper 2's effect estimates.

4. Step 0: shared foundation (months 1–6)

Step 0 builds the pieces every paper uses, once, with tests. Nothing in Step 0 looks at a buffering result.

S0.1 Two-scale grid and crosswalk	
Goal	One table that maps every 30 m pixel to its 480 m block and its 1/24° cell, with area weights.
Inputs	Landsat / OpenET tile frames (frames.json), lab gridMET grid, hp_bound2010 polygon.
Method	Pixel centres projected exactly with pyproj (GDAL nearest-neighbour warping is approximate; this was measured in the CDL work). Area weights for cells that cross UTM zones; aquifer fraction per block and cell; block statistics (mean, SD, count, class fractions).
Output	Crosswalk tables (Parquet); CF-compliant block grid definitions; aggregation function with tests.
Verify	(1) Each pixel maps to exactly one block per zone. (2) Block means equal an independent recomputation to float tolerance on three random tiles. (3) Summed pixel area per cell equals cell area × aquifer fraction within 0.5 %.
Depends on	Nothing.

S0.2 Data completion and one quality layer	
Goal	All inputs needed by Paper 1 exist for the focus landscapes, then for the aquifer; one module applies every quality rule.
Inputs	Landsat 30 m and OpenET v2.1 tiles, CDL, OCO-2 (export ends 2025-09-02; finish to present), OCO-3, ECOSTRESS v002 native tiles, SMAP L3/L4.

S0.2 Data completion and one quality layer	
Method	Rules: Landsat pixel QA and aerosol screen as in the notebook; OCO quality flag, and the undeclared value -4.637617×10^{12} in *Science_SIF_Unadjusted_771nm* treated as missing; SMAP retrieval-quality bit 0; DEM zero block masked; days the Climatology Lab flags dropped. Landsat 5/7 to 8/9 continuity (Roy et al. 2016) follows the notebook's harmonisation switch; headline results are repeated with it off.
Output	Per-file manifest (variables, period, valid fraction inside the aquifer); quality module.
Verify	(1) Manifest regenerated by script agrees with the 2026-10-20 audit. (2) Each quality rule has a unit test with a planted defect that it catches. (3) Focus-landscape tiles complete for 2008–2025 before aquifer-wide processing starts.
Depends on	S0.1 for tile lists.

S0.3 Supply and demand drivers at both scales	
Goal	Monthly drivers with no time leakage and no saturation of drought indices.
Inputs	gridMET daily; SMAP L4; gSSURGO available water storage.
Method	Rainfall accumulated over 30, 60, 90, 180 and 365 days ending before the month; in-house SPI and SPEI fitted per cell and calendar month on 1985–2014 (Stagge et al. 2015; Vicente-Serrano et al. 2010), continuous beyond the lab files' ±2.09 cap; EDDI for evaporative demand (Hobbins et al. 2016); VPD, radiation, temperature. A one-layer soil-water bucket driven by gridMET with capacity from gSSURGO supplies root-zone wetness for 2000–2014, when SMAP does not exist.
Output	Driver cubes at scales C and F; bucket wetness 1985–2025.
Verify	(1) SPI has $\lvert\text{mean}\rvert < 0.05$ and $\lvert\text{SD} - 1\rvert < 0.05$ per cell on the baseline. (2) Spearman correlation with the lab spi90d ≥ 0.9. (3) EDDI correlates negatively with SPEI. (4) Bucket wetness versus SMAP L4 root zone, 2015–2024: correlation of anomalies reported per vegetation class.
Depends on	S0.1.

S0.4 Dry-down event catalogue	
Goal	One table of drought events per cell and block, used by all three papers.
Method	Onset when SPI-90 ≤ −0.8 (US Drought Monitor D1 (Svoboda et al. 2002)) for at least two consecutive months; end when SPI-90 > −0.5 for two months. Flash-drought variant from rapid soil-moisture percentile decline (Otkin et al. 2018). Attributes: onset, peak, duration, severity, demand anomaly, season.
Output	Event table; event-membership flag per cell-month.
Verify	(1) 2012 and 2022 events present in ≥ 80 % of aquifer cells. (2) Monthly drought area agrees with US Drought Monitor D1+ area (correlation ≥ 0.7). (3) Catalogue rebuilt with SPEI and with a −1.0 threshold; overlap reported.
Depends on	S0.3.

S0.5 Groundwater exposure and dynamics	
Goal	Static depth to water at both scales and a dynamic water-level signal without interpolation.
Inputs	HRES-WTD 30 m (Ma et al. 2026); well time series (USGS, NECSD, TWDB); state records to acquire (Section 10).
Method	Per block: mean, SD and fraction of pixels with depth to water below 2, 5 and 10 m. Per well: annual anomaly, 2000–2025 trend (Sen slope), rebound episodes (rise above measurement noise sustained for two years). Aggregated to the 0.25° well clusters of the existing cluster script.
Output	Exposure layers; well-cluster dynamics table; list of rebound clusters.
Verify	(1) Wells per year and state reported; USGS counts fall from about 7,400 to about 3,300 a year after 2015 while NECSD and TWDB continue, so coverage gaps are explicit. (2) Trend sign agrees with the USGS 2017–2019 change map in ≥ 80 % of cells with ≥ 3 wells. (3) HRES-WTD versus measured 2015 depth: bias and RMSE by depth class.
Depends on	S0.1; acquisitions for 2016+ in Kansas and Colorado.

S0.6 Terrain position and stream layers	
Goal	Topographic exposure that is independent of the vegetation signal.
Method	Height above nearest drainage (Nobre et al. 2011) from GLO-30 with NHDPlus HR flowlines; slope; topographic wetness; distance to stream; mapped alluvium.
Verify	(1) ≥ 95 % of channel pixels have HAND < 1 m. (2) HAND versus HRES-WTD rank correlation reported. (3) The zero-filled DEM block is excluded.
Depends on	S0.1; flowlines and alluvium (Section 10).

S0.7 Registered analysis plan and synthetic test harness	
Goal	Fix hypotheses and tests in advance, and prove that each pipeline recovers effects of known size.
Method	A dated, version-controlled document with hypotheses, outcomes, pass criteria and robustness checks for Papers 1–3. A harness of synthetic cubes on the real grid and calendar with planted effects: a subsidy of known size, a threshold at a known depth, a known lag between signals.
Verify	Planted effects recovered: subsidy amplitude bias < 10 %; threshold within ±1 m; lag within ±1 month. The harness runs in continuous integration.
Depends on	S0.1–S0.6.

Gate G0 (month 6)
Crosswalk, quality layer, drivers and event catalogue pass their tests. If the 480 m aquifer-wide build is not ready, Paper 1 starts in the focus landscapes and at scale C, and the aquifer-wide fine product follows.

5. Paper 1: a dynamic, vegetation-based map of water subsidy (months 4–16)

Working title: *When does vegetation use water that rain did not supply? A dynamic, ET- and canopy-based map of water subsidy across the High Plains aquifer.*

5.1 Rationale

Irrigation maps for the region are annual: AIM-HPA (Deines et al. 2019a), LANID (Xie et al. 2021) and IrrMapper (Ketchum et al. 2020) label a field once per year, and the products in the repository (GFSAD, the global irrigation map to 2015) are single-epoch or coarse. For buffering, the question is not whether a pivot exists but whether, in a given month, the vegetation is drawing on water beyond rainfall. That is a biological signal: canopy water and ET that stay high while comparable rain-fed vegetation declines. Lowry & Loheide (2010) named the extra water a *groundwater subsidy*; satellite studies of groundwater-dependent vegetation read it from greenness and ET persistence (Eamus et al. 2015; Orellana et al. 2012; Huntington et al. 2016; Gou et al. 2015). Dryland ecology gives the expected rain-limited behaviour: production responds to rainfall pulses (Noy-Meir 1973) and scales with rainfall along a rain-use-efficiency curve (Huxman et al. 2004; Knapp & Smith 2001), and departures from that curve are the basis of residual-trend methods (Evans & Geerken 2004; Burrell et al. 2017).

Paper 1 turns this into a monthly state for every pixel: rain-coupled, subsidised, stressed, bare or flooded. The state comes from the vegetation's own departure from a rain-only expectation. Hydrological and administrative data (depth to water, wells, pumping records, irrigation maps) are used to label and validate the states, not to make them. The same departure, in millimetres of ET, gives a monthly volume of subsidised water.

5.2 Why the expected-function model is rebuilt

- **Training population.** A model fitted on all pixels learns subsidised behaviour through its proxies (an irrigated cell has a high summer ET normal, particular soils, particular crops) and so absorbs part of the signal it should expose. Here the model is trained only on vegetation that can use nothing but rain, so its prediction has a defined meaning: what rain-only vegetation of this type, on this soil, with this weather history would do this month.

- **Target.** ET alone misses the plant's own state. The new model predicts ET, greenness (NIRv (Badgley et al. 2017)) and canopy water (NDMI (Gao 1996)) jointly; a subsidised canopy is green, wet and transpiring at once, and canopy temperature from ECOSTRESS adds an energy check where it exists.
- **Transfer.** Applying a model trained on deep, upland, rain-fed pixels to valleys and irrigated fields is extrapolation. Its area of applicability is computed per pixel-month and results outside it are withheld (Meyer & Pebesma 2021).
- **Independence caveat.** OpenET models use Landsat surface temperature and reflectance, so ET and the Landsat indices share inputs. Agreement between them is partly built in; ECOSTRESS and SMAP VOD are the independent checks.

Questions and hypotheses (Paper 1)

- **Q1.1** Can a monthly water-subsidy state be detected from vegetation and ET alone, with stated accuracy?
- **Q1.2** Can the subsidy be attributed to irrigation, natural groundwater or surface water from its biological and spatial signature?
- **Q1.3** How much does a dynamic mask change where and when subsidised water use is found, compared with static masks?
- **H1.1** In drought months the subsidy state separates irrigated from rain-fed fields with balanced accuracy ≥ 0.85 against pumping records and AIM-HPA.
- **H1.2** Natural subsidy is most frequent where the water table lies within reach of roots: its probability falls with depth to water and HAND (Fan et al. 2017; Kollet & Maxwell 2008).
- **H1.3** A static annual mask misclassifies at least 10 % of irrigated area-months, concentrated in drought years and in areas changing management.

5.3 Steps

P1.1 Reference population: vegetation that can use only rain	
Goal	A dynamic, year-by-year set of pixels whose only water source is rain.
Inputs	HRES-WTD, HAND (S0.6), GDE map, NWI wetlands, water indices, CDL class, registered wells and points of diversion (Section 10).
Method	A pixel-year is reference when all hold: depth to water > 30 m everywhere in its block (sensitivity 15 and 60 m); HAND > 10 m; outside GDE and wetland polygons and never open water that year; no registered irrigation well or point of diversion within 800 m; CDL class is grassland, shrubland or a crop grown dryland in that county. These rules use only exposure and administrative data, not the vegetation signal, so the selection is not circular.
Output	Yearly reference mask; counts by climate stratum (aridity decile) × vegetation class.
Verify	(1) Every aridity decile × vegetation class has ≥ 500 reference blocks, or is declared out of domain. (2) Plausibility: water-year ET ≤ rainfall + soil storage change + 10 % in ≥ 95 % of reference pixel-years (a diagnostic, not a selection rule). (3) Reference maps by year published.
Depends on	S0.1, S0.5, S0.6.

P1.2 Rain-only expectation model	
Goal	For each pixel-month, the distribution of ET, NIRv and NDMI that rain-only vegetation would show.
Inputs	No exposure variables (rule R2). Twelve months of driver history (S0.3); soil available water and productivity (gSSURGO); elevation; vegetation class (CDL; MODIS land cover before 2008); a phenological clock from the class's reference calendar (Jönsson & Eklundh 2004); pre-season state (NIRv and NDMI of the three months before the season), lagged so that the current season cannot leak in.
Method	**M1 (primary):** a multi-output probabilistic sequence model (temporal convolution or LSTM encoder of the driver history, static encoder, quantile heads per target) trained only on reference pixel-months, with a water-year mass penalty (ET may not exceed rain plus storage change), combining data-driven learning with a physical constraint (Reichstein et al. 2019). Intervals calibrated by conformalised quantile regression (Romano et al. 2019). **M2 (ecological benchmark):** rain-use-efficiency / residual-trend curves per class and climate stratum, with the rainfall window chosen on reference data (Evans & Geerken 2004; Burrell et al. 2017). **M0 (floor):** per-pixel monthly climatology.
Output	Median and 5–95 % quantiles of each target per pixel-month at scales F and C.

P1.2 Rain-only expectation model

Verify	(1) Cross-validated on reference data with spatial × year blocks: M1 beats M0 on anomaly RMSE in every climate stratum and beats M2 overall. (2) 90 % interval coverage between 0.88 and 0.92 on held-out reference blocks. (3) On the synthetic harness a planted subsidy in 10 % of pixels is recovered with bias < 10 %. (4) A known-zero set (dryland fields with no water right, deep water table, but HAND < 10 m, so outside the reference by one rule) shows no excess beyond the 5 % false-positive rate.
Depends on	P1.1, S0.3.

P1.3 Transfer check: area of applicability

Goal	Know where the expectation model is extrapolating.
Method	Dissimilarity index and area of applicability per pixel-month (Meyer & Pebesma 2021); a classifier that tries to tell reference from target pixels (its accuracy per stratum measures covariate shift); an importance-weighted refit as a sensitivity.
Output	Applicability flag per pixel-month; summary by stratum and focus landscape.
Verify	(1) Share of target pixel-months inside the area of applicability reported per stratum. (2) All later statistics restricted to it. (3) If a focus landscape has < 60 % inside, its natural-vegetation analysis in Paper 2 uses local donor designs (P2.3–P2.4) only.
Depends on	P1.2.

P1.4 Subsidy index and ET excess

Goal	A per-pixel-month measure of how far vegetation exceeds its rain-only expectation, and the water volume involved.
Method	Standardised departure per target, z = (observed − median) / interval half-width. Joint subsidy index SI = signed Mahalanobis distance of the departure vector, using the covariance of reference residuals (positive when ET and canopy water both exceed expectation). ET excess = max(0, ET − predicted median) in mm, with interval.
Output	SI and ET excess per pixel-month; aggregated at F and C.
Verify	(1) On held-out reference blocks SI follows its reference distribution: false-positive rate at the 95 % threshold within 4–6 %. (2) Irrigated versus rain-fed tower pair US-Ne1 / US-Ne3 near Mead, Nebraska (Suyker & Verma 2009): the irrigated site exceeds the threshold in dry months and the rain-fed site does not (check first whether the towers fall inside the aquifer polygon). (3) ET excess compared with tower ET minus rainfall at all AmeriFlux sites in the region.
Depends on	P1.2, P1.3.

P1.5 Dynamic monthly states (the mask)

Goal	A probabilistic monthly state per pixel: rain-coupled (R), subsidised (S), stressed (X), bare or harvested (B), open water or flooded (W).
Method	Hidden Markov model per vegetation class (Rabiner 1989) with month-specific transition matrices; emissions are SI, the three z-scores, NIRv level and water indices (MNDWI, AWEI, WI2015 (Xu 2006; Feyisa et al. 2014; Fisher et al. 2016)). Posterior state probabilities form the dynamic mask; the most likely path gives maps. As an independent check, a Bayesian change-point decomposition (Zhao et al. 2019) of each pixel's NDMI and ET series dates the onset and end of subsidy within a season; BFAST and CCDC (Verbesselt et al. 2010; Zhu & Woodcock 2014) are fallbacks.
Output	Monthly state probabilities 2008–2025 at 30 m (focus landscapes) and 480 m fractions (aquifer).
Verify	(1) Onset months from the hidden Markov model and from the change-point method agree within ±1 month for ≥ 70 % of irrigated pixel-seasons. (2) State durations plausible by crop (for example, irrigated maize subsidised through mid-summer). (3) Maps of posterior uncertainty (entropy) published with the states.
Depends on	P1.4.

P1.6 Source attribution: irrigation, natural groundwater or surface water	
Goal	Label each subsidised pixel-month with its likely water source, or 'unresolved'.
Method	Features from the biological and spatial signature: sharpness of the subsidy at field edges, timing against the crop calendar, abruptness of onset (irrigation starts suddenly; natural subsidy follows the water table), synchrony with neighbouring fields, persistence across years, coincidence with open water, and HAND and depth to water (allowed here because this step attributes the source; it does not build the expectation). Classifier: gradient boosting with spatially blocked cross-validation. Labels: irrigation from Kansas reported water use by point of diversion, Nebraska certified irrigated acres, AIM-HPA and LANID; natural groundwater from the GDE map (Rohde et al. 2024b), riparian NWI and wet meadows outside any irrigated place of use, following the multi-evidence logic of continental GDE mapping (Doody et al. 2017); surface water from mapped playas with water-index flooding.
Output	Source label and probability per subsidised pixel-month.
Verify	(1) Balanced accuracy by class and by year type (dry, normal, wet) with 95 % intervals; irrigation in drought years ≥ 0.85 (H1.1). (2) Precision of the natural-groundwater class ≥ 0.7 against GDE labels; otherwise that class is reported as 'unresolved subsidy' (Gate G1).
Depends on	P1.5; labels (Section 10).

P1.7 Volumes against pumping, and dynamic versus static masks	
Goal	Check the water volume implied by the subsidy, and quantify what the dynamic mask adds.
Method	ET excess summed per place of use and year against reported pumped volume in Kansas, following the comparisons of Filippelli et al. (2022) and Majumdar et al. (2020). Dynamic irrigated area-months compared with AIM-HPA and LANID by year and with LEMA status.
Verify	(1) Slope, R^2 and the ratio of ET excess to pumping with intervals; a ratio below 1 is expected because of conveyance and deep percolation, and the implied efficiency is reported. (2) H1.3 tested by the share of area-months on which the dynamic and static masks disagree.
Depends on	P1.4–P1.6; Kansas water-use data.

Deliverables and Gate G1 (month 16)

- Monthly state probabilities and ET excess, 2008–2025: 480 m aquifer-wide and 30 m in the focus landscapes; code and data release; Paper 1 submitted.
- Gate G1: if the natural-groundwater class fails its precision test, Paper 2's natural thread relies on the design-based contrasts (P2.3–P2.4) and reports subsidy without a source label.

6. Paper 2: buffering effect and sustainability (months 14–28)

Working title: *Groundwater access and the persistence of vegetation function through drought in the High Plains: effect, limits and sustainability.*

6.1 Rationale

Shallow groundwater can supply root-zone water during dry spells (Lowry & Loheide 2010; Miguez-Macho & Fan 2021), groundwater and vegetation covary spatially across the globe (Koirala et al. 2017), rooting depth itself follows the water table (Fan et al. 2017), and land-surface energy fluxes are most sensitive to groundwater within a critical depth range (Kollet & Maxwell 2008; Maxwell & Condon 2016). Isotope studies show that plant use of groundwater is common (Evaristo & McDonnell 2017), and groundwater-dependent ecosystems are widespread in drylands (Rohde et al. 2024b). In the High Plains, groundwater decline has already been linked to the loss of stream fish assemblages (Perkin et al. 2017). Satellite gravimetry shows storage declines in heavily irrigated regions (Rodell et al. 2018). What is missing is a causal estimate of how much access to groundwater keeps vegetation functioning through drought, separated from climate, soils, topography and pumping, and a test of whether that buffering is paid for by depletion (Cuthbert et al. 2023; Thomas et al. 2017).

Questions, hypotheses and outcomes (Paper 2)

- **H2.1** Within a climate stratum, natural and rain-fed vegetation with the water table within root reach retains more function during dry-downs than matched vegetation without access; the effect is near zero in wet periods and declines with depth to water.
- **H2.2** The effect is larger for deep-rooted woody and wet-meadow vegetation than for grassland and rain-fed crops.
- **H2.3** Where wells decline, irrigation-subsidised persistence is volumetrically consistent with the storage loss (mined); where wells are stable it is not.
- **Outcomes:** O1 event-mean subsidy index SI (Paper 1); O2 function retention, NIRv or ET at the event peak divided by the same-season pre-event baseline; O3 canopy-water retention (NDMI); secondary, SIF retention at scale C.

6.2 Steps

P2.1 Exposure at event onset

Goal	A definition of groundwater access that uses only information available before each event.
Method	Access = 1 when the block-median depth to water minus the rooting depth of the vegetation class (Fan et al. 2017; Stocker et al. 2023) is ≤ 2 m at onset; the continuous depth is also kept. Dynamic depth = HRES-WTD 2015 baseline + well-cluster anomaly (S0.5). The cluster map (Figure 1) is a reporting stratum.
Verify	(1) A time-stamp test proves that no post-onset value enters. (2) Count table of exposed and unexposed units in every climate stratum; strata without both are dropped and listed.
Depends on	S0.5, P1.5 (to exclude irrigated units from the natural thread).

P2.2 Design A: matching inside climate strata

Goal	The average and heterogeneous effect of access on O1–O3.
Method	Strata = aridity × reference-ET deciles. Inside each, exposed and unexposed blocks of the same vegetation class are matched (Mahalanobis distance with calipers) on soil water capacity, elevation, slope, pre-event three-year function mean and trend, and event severity (Stuart 2010; Rosenbaum & Rubin 1983). Effects by augmented inverse-probability weighting with cross-fitted nuisance models (Chernozhukov et al. 2018); heterogeneity with causal forests (Wager & Athey 2018; Athey et al. 2019); trimming to common support (Crump et al. 2009).
Verify	(1) Standardised mean differences < 0.1 for every matching variable after matching. (2) The synthetic harness recovers a planted effect within its interval. (3) Wet-period placebo (SPI-90 ≥ +0.8) effect indistinguishable from zero.
Depends on	P2.1, P1.4.

P2.3 Design B: valley–upland transects and an alluvial-boundary discontinuity

Goal	An estimate that does not rely on a matching model, inside one climate.
Method	In FL1, FL3, FL4 and FL7, transects perpendicular to the channel: outcome against HAND and depth to water. A spatial regression discontinuity at the mapped alluvium edge (Keele & Titiunik 2015), where depth to water changes abruptly but climate does not.
Verify	(1) First stage: depth to water changes significantly at the boundary. (2) Covariates (soils, elevation trend, pre-event function) continuous across it. (3) Estimates stable across three bandwidths.
Depends on	S0.6, P1.4; alluvium maps.

P2.4 Design C: within-unit difference-in-differences across events

Goal	Remove time-invariant confounding (topography, soils) by comparing each unit with itself.
Method	Outcome in dry-down months minus non-drought months of the same season, exposed minus unexposed; event-study plot with leads and lags around onset. For units where P1.3 flagged poor transfer, donor-weighted counterfactuals (generalised synthetic control (Abadie et al. 2010; Xu 2017)) from unexposed blocks of the same class and stratum.
Verify	(1) Pre-onset leads indistinguishable from zero. (2) Donor pre-period fit reported; units with poor pre-fit excluded and counted.

P2.4 Design C: within-unit difference-in-differences across events	
Depends on	S0.4, P1.4.

P2.5 Design D: the Sheridan-6 LEMA as a management experiment	
Goal	Separate the effect of management from that of groundwater on the irrigation thread.
Method	Difference-in-differences inside versus comparable land outside the LEMA, before and after 2013, for irrigation subsidy (ET excess), crop mix and well levels (Deines et al. 2019b; Drysdale & Hendricks 2018; Deines et al. 2021).
Verify	(1) Parallel pre-trends 2005–2012. (2) The direction of the pumping change agrees with the published estimates.
Depends on	P1.4–P1.7.

P2.6 Robustness battery (pre-specified, reported in one panel)	
Method	Wet-period placebo; negative-control outcome (dormant-season NIRv of grassland, and pre-event outcome leads (Lipsitch et al. 2010)); E-values (VanderWeele & Ding 2017); HAND-class stratification; leave-one-landscape-out; replication at 30 m, 480 m, 960 m and 1/24°; drought definition replicated with SPEI and flash drought; added spatial smooths at several ranges to probe spatial confounding (Paciorek 2010); comparison of designs A–D (Butsic et al. 2017).
Verify	Every check is reported, including those that fail. A design that fails its own diagnostic is reported but not used for the headline.
Depends on	P2.2–P2.5.

P2.7 Sustainability classification with volumes	
Goal	Classify buffered systems as sustainable, mined, irrigation-subsidised, failed buffer or no buffer, with probabilities.
Method	Per well cluster and management district: annual ET excess volume (natural and irrigated, Paper 1) against (i) reported pumping, (ii) storage change = water-level change × specific yield (gridded; 0.10–0.20 as sensitivity), (iii) GRACE groundwater storage anomaly (total storage minus SMAP soil moisture) at basin scale (Scanlon et al. 2012; Watkins et al. 2015). The original volume-plausibility rule: 'mined' only when persistent function, declining wells and a volume ratio within a stated factor coincide. Depletion projections for Kansas (Steward et al. 2013); water-balance framing from Butler et al. (2016) and Whittemore et al. (2016); renewability from Cuthbert et al. (2023).
Verify	(1) Every water-balance term carries an interval. (2) Share of units that change class across the specific-yield range reported. (3) Class map compared with the Kansas water-balance results of the cited studies where they overlap.
Depends on	P1.7, S0.5; specific-yield grid.

Deliverables and Gate G2 (month 24)

- Effect maps with intervals at both scales; effect by vegetation class and depth; sustainability class map with probabilities; Paper 2 submitted at month 28.
- Gate G2: if designs A–C disagree in sign for the natural thread, the result is reported as unresolved and the failing assumption is identified before Paper 3 fits thresholds to it.

7. Paper 3: thresholds, reversibility and mechanism (months 22–34)

Working title: *Where groundwater buffering fails, whether it returns, and how failure unfolds: thresholds, hysteresis and water–carbon–energy decoupling in the High Plains.*

7.1 Rationale

Management targets for groundwater-dependent ecosystems need thresholds (Rohde et al. 2024a; Saito et al. 2021), but ecological thresholds and regime shifts (Scheffer et al. 2001) are often asserted rather than tested against smooth alternatives (Groffman et al. 2006; Andersen et al. 2009). Riparian trees decline after alluvial water-table drops (Scott et al. 1999; Naumburg et al. 2005; Rohde et al. 2021), yet whether function returns when water returns is open: hydrological systems can fail to recover after drought (Peterson et al. 2021; Saft et al. 2015), forests carry drought legacies (Anderegg et al. 2015), and recovery time varies widely (Schwalm et al. 2017). Groundwater recovery itself is documented only in some aquifers (Jasechko et al. 2024). Near failure, the order in which canopy water, photosynthesis and ET decline should depend on plant hydraulic strategy (Martínez-Vilalta et al. 2014; Konings & Gentine 2017) and on whether demand or supply limits the plant (Novick et al. 2016).

Hypotheses (Paper 3)

- **H3.1** Buffering declines with depth to water; for groundwater-dependent natural vegetation a threshold model fits better than a smooth one.
- **H3.2** Where water levels rebound, herbaceous function returns; woody riparian function shows hysteresis.
- **H3.3** Near failure, canopy water and SIF decline before ET in more isohydric vegetation; ET persists while canopy water falls in more anisohydric vegetation.
- **H3.4** Decoupling among the signals increases as the distance to the failure threshold shrinks.

7.2 Part A: thresholds (RQ2)

P3.1 Response curves and threshold tests	
Goal	Where buffering weakens, and whether it does so abruptly.
Method	Event-level buffering (Paper 2 effect and SI) against depth to water, its trend, storage anomaly, vegetation class, rooting depth and drought severity. Segmented regression with unknown break (Muggeo 2003; Toms & Lesperance 2003); shape-constrained additive models (Pya & Wood 2015); hierarchical Bayesian change-point by vegetation class. Threshold and smooth models compared by leave-one-out cross-validation (Vehtari et al. 2017).
Verify	(1) The harness recovers a planted break within ±1 m at the real sample size. (2) Break location reported with its credible interval. (3) 'Threshold' is used only when the threshold model wins by more than two standard errors of the cross-validation difference.
Depends on	P2.2–P2.6.

P3.2 Distance-to-failure map	
Method	Per block: (current depth to water − threshold) / its uncertainty; with the well trend, years to threshold at the current rate.
Verify	Hindcast: thresholds fitted on 2005–2015 predict where function loss appears in 2016–2025; skill against a depth-only baseline reported.
Depends on	P3.1, S0.5.

7.3 Part B: reversibility (RQ3)

P3.3 Rebound selection	
Goal	Test recovery only where water actually returned (original plan rule).
Method	Well clusters with a significant rise (S0.5), expected mainly in Nebraska after 2018–2019 and in parts of Texas, where records continue; stream reaches with returning flow.
Verify	List of rebound clusters with rise and uncertainty. With fewer than 10 clusters the analysis is framed as case studies.
Depends on	S0.5; state well records.

P3.4 Function–water-level loops	
Method	For each rebound cluster and a same-climate control: trajectories of function residuals (from the rain-only model, so wet years do not mimic recovery) against water level, using MODIS 2000–, Landsat 2005–, OpenET 2000– and GOSIF as context. Loop area and direction by a hysteresis index (Zuecco et al. 2016). Classes from the original plan: reversible decline, hysteretic decline, water never returned.
Verify	(1) Hysteresis claimed only when the index interval excludes zero. (2) Same analysis on controls shows no loop.
Depends on	P3.3, P1.2.

P3.5 Recovery time and resilience indicators (secondary)	
Method	Recovery time after events (Schwalm et al. 2017); rolling lag-1 autocorrelation and variance of deseasonalised NIRv residuals (Dakos et al. 2012; Verbesselt et al. 2016; Smith et al. 2022; Forzieri et al. 2022), only where at least two decline–recovery cycles exist (Scheffer et al. 2009).
Verify	Significance against phase-randomised surrogate series; reported as secondary evidence.
Depends on	P1.2, P3.3.

7.4 Part C: mechanism (RQ4)

P3.6 Sequence of decline within dry-downs	
Goal	The order and lags in which canopy water, ET, photosynthesis and canopy temperature depart from normal.
Method	Space-for-time across many events and locations (Blois et al. 2013), with its limits stated (Damgaard 2019). Per event and cell: onset of decline of NDMI and VOD (water), OpenET and ECOSTRESS ET (energy and water flux), OCO-2/3 SIF pooled per cell-event (carbon (Porcar-Castell et al. 2021)), and ECOSTRESS canopy minus air temperature, dated by change-point detection (Zhao et al. 2019). Coupling by wavelet coherence (Grinsted et al. 2004) and lagged information flow (Ruddell & Kumar 2009; Runge et al. 2019).
Verify	(1) Planted lags recovered within one time step at observed noise. (2) Soundings per cell-event reported; cells below a minimum are excluded. (3) OCO-2 and OCO-3 agree where both sample.
Depends on	S0.4, P1.2.

P3.7 Plant hydraulic strategy from microwave canopy water	
Method	Ecosystem isohydricity from SMAP VOD dynamics, including the AM–PM difference (Konings & Gentine 2017; Konings et al. 2019). Because VOD responds to both biomass and leaf water potential (Momen et al. 2017), GEDI biomass is a covariate and the diurnal difference is the primary statistic.
Verify	Isohydricity by vegetation class with intervals; its relation to the decline order of P3.6 tested against H3.3.
Depends on	P3.6.

P3.8 Synthesis: decoupling against distance to failure	
Method	Decoupling metric D (loss of coherence, growth of lag) modelled as a function of distance to failure, water-level trend, drought intensity, vegetation class and subsidy state.
Verify	H3.4 tested with the same threshold-versus-smooth comparison as P3.1.
Depends on	P3.2, P3.6, P3.7.

Deliverables and Gate G3 (month 30)

- Distance-to-failure map; reversibility class for each rebound cluster; decline sequence by vegetation class; Paper 3 submitted at month 34.
- Gate G3: with fewer than 10 rebound clusters, Part B is written as case studies and Paper 3 leads with thresholds and mechanism.

8. Additional analyses

Each is optional and sized to fit inside a paper if time allows.

ID	Analysis	Data	Verification	Fits in
A1	Hydroperiod of Sandhills lakes and playas from monthly water indices, as an indicator of groundwater (lakes) versus rain-fed recharge features (playas (Gurdak & Roe 2010))	MNDWI, AWEI, WI2015; wells	Agreement with Global Surface Water occurrence (Pekel et al. 2016); lake area versus nearby well level	Paper 1 (state W), Paper 3
A2	Riparian function versus stream intermittence and stream–aquifer disconnection (Jasechko et al. 2021; Perkin et al. 2017)	NWIS discharge, NHDPlus, NIRv	Event-study around flow cessation	Paper 2 or 3
A3	Buffering in yield terms: county yields and field yield maps (Lobell et al. 2015); shallow groundwater and yield (Zipper et al. 2015)	NASS yields, Landsat	Out-of-sample yield prediction	Paper 2
A4	Scale study: effect estimates at 30 m, 480 m, 960 m, 1/24° and 0.25°	All	Sign and magnitude by scale	Methods note
A5	Woody structure as a modifier of buffering	GEDI L4B	Heterogeneity in P2.2	Paper 2
A6	Independent ET reference: comparison with GLEAM4 groundwater-related evaporation (Miralles et al. 2025; Hulsman et al. 2023)	GLEAM4	Spatial agreement of buffered areas	Paper 2

9. Timeline and decision gates



Figure 3. Thirty-six-month timeline. Triangles mark paper submissions (months 16, 28 and 34); diamonds mark decision gates. Bars are labelled directly; colours match the step cards.

Gate	Month	Test	If it fails
G0	6	Crosswalk, quality layer, drivers and event catalogue pass (S0.1–S0.4, S0.7)	Paper 1 starts in focus landscapes and at scale C
G1	16	Reference model skill and calibration (P1.2); natural-groundwater precision ≥ 0.7 (P1.6)	Natural thread of Paper 2 uses design-based contrasts only; subsidy reported without source

Gate	Month	Test	If it fails
G2	24	Designs A–C agree in sign; robustness panel complete (P2.6)	Report as unresolved; diagnose the failing assumption before thresholds
G3	30	At least 10 rebound clusters (P3.3)	Reversibility as case studies

10. Data to acquire

Only what unblocks a step, in priority order.

#	Dataset	Unblocks	Note
1	Kansas water rights, points of diversion and reported annual water use (WIMAS)	P1.1, P1.6, P1.7, P2.5	Field-scale pumping volumes and irrigation labels
2	Nebraska registered wells; NRD certified irrigated acres and metered pumping where available	P1.1, P1.6	Labels in the north
3	AIM-HPA annual irrigation maps (Deines et al. 2019a) and LANID (Xie et al. 2021)	P1.6, P1.7	Validation only
4	Kansas WIZARD and Colorado DWR water levels, 2016 onward	S0.5, P3.3	USGS records thin out after 2015 in these states
5	Gridded specific yield and saturated thickness for the aquifer (USGS)	P2.7	Replaces the uniform 0.15
6	NHDPlus HR flowlines, NWIS daily discharge, mapped alluvium	S0.6, P2.3, A2	Terrain and boundary designs
7	Playa layer (Playa Lakes Joint Venture) and National Wetlands Inventory	P1.1, P1.6, A1	Natural and surface-water labels
8	US Drought Monitor weekly maps	S0.4	Event-catalogue check
9	Rooting-depth map (Fan et al. 2017)	P2.1	Exposure definition
10	NASS county yields; optional Landsat GPP (Robinson et al. 2018)	A3	Outcomes

11. Risks and mitigations

Risk	Effect	Mitigation
Reference population thin in the wetter north	Expectation model extrapolates there	Area of applicability (P1.3); local donor designs (P2.4); 15 m depth sensitivity
Growing-season cloud gaps in Landsat	Missing or biased monthly composites	Observation counts carried; minimum-count rule; OpenET as a gap-tolerant ET signal
Disagreement among OpenET models	ET excess uncertain	Ensemble with model spread as uncertainty; key results repeated with two single models
Well records thin after 2015 in Kansas, Colorado and Oklahoma	Reversibility limited to Nebraska and Texas	State acquisitions (Section 10); Gate G3
Sparse SIF soundings	Carbon leg noisy	Pool by cell and event at scale C; GOSIF not used as independent evidence
VOD mixes biomass and water	Hydraulic strategy misread	Diurnal difference; GEDI biomass covariate
Storage and compute (about 3.5 TB of Landsat for 2005 onward)	Delays	Focus landscapes first; tile-by-tile 480 m build; Gate G0 fallback
Spatial confounding by topography and climate	Biased effects	Designs B–D, E-values, scale replication

Risk	Effect	Mitigation
Many strata and outcomes	False discoveries	Registered plan; false-discovery-rate control

12. Future work beyond the dissertation

- Transfer the framework to the Central Valley and other declining aquifers listed in the original plan, once the High Plains methods are stable.
- The original 'advanced extension': a differentiable land-surface model that learns a spatial field of accessible storage, now trained against the rain-only residual of Paper 1. An identifiability test comes first: the learned field must be stable across events and sensors.
- An isotope and sap-flow campaign in FL1 and FL2 to confirm the water source of subsidised vegetation (Evaristo & McDonnell 2017; Miguez-Macho & Fan 2021).
- A decision-support product for groundwater management districts combining distance to failure, sustainability class and dynamic subsidy (Rohde et al. 2024a; Saito et al. 2021).
- Biodiversity outcomes of groundwater dependence (Saccò et al. 2024; Perkin et al. 2017).

13. Open questions for you

- **Focus landscapes.** Keep FL1–FL7, or replace or add sites (for example field sites your group already instruments)?
- **Reference rules.** Are 30 m depth to water and 10 m HAND sensible for the High Plains, or do you have local knowledge of deeper rooting (for example, in the Sandhills)?
- **Pumping and label data.** Does the lab have contacts or agreements for Kansas WIMAS, Nebraska NRD meter data or Colorado DWR records?
- **Data paper.** Should the Paper 1 product also go out as a data descriptor?
- **Compute.** Will the aquifer-wide 30 m to 480 m processing run on Colab and Drive, or on a lab server?
- **Canopy temperature.** Should ECOSTRESS canopy temperature be a target of the rain-only model (only 2018 onward), or used only in Paper 3?
- **Season definition.** One growing season (April–October) for all classes, or class-specific calendars (winter wheat differs)?
- **Journals.** Any preferred venues per paper?

Appendix A. The 18 Landsat indices and their roles

Formulas are those implemented in the Landsat 30 m notebook; the sources below define them. To limit multiple testing, the registered plan names a primary set (NIRv, NDMI, MNDWI, plus OpenET ET) and treats the rest as sensitivity checks.

Index	Family	Source	Role in this plan
NIRv	Greenness	(Badgley et al. 2017)	Primary greenness and photosynthesis proxy (P1.2, P3.6)
kNDVI	Greenness	(Camps-Valls et al. 2021)	Saturation-robust check over dense irrigated canopies
NDVI	Greenness	(Tucker 1979)	Comparability with MODIS and earlier work
EVI, EVI2	Greenness	(Huete et al. 2002; Jiang et al. 2008)	Long-record consistency (EVI2 needs no blue band)

Index	Family	Source	Role in this plan
SAVI, MSAVI, OSAVI	Greenness, soil-adjusted	(Huete 1988; Qi et al. 1994; Rondeaux et al. 1996)	Sparse rangeland and early-season canopies
GNDVI	Greenness (chlorophyll)	(Gitelson et al. 1996)	Chlorophyll-sensitive check
NDMI	Canopy water	(Gao 1996)	Primary canopy-water signal (P1.2, P3.6)
NMDI	Canopy and soil water	(Wang & Qu 2007)	Separates soil from canopy drying
MSI	Canopy water (inverse)	(Hunt & Rock 1989)	Sensitivity check
GVMI	Canopy water	(Ceccato et al. 2002)	Sensitivity check
NDWI (McFeeters)	Open water	(McFeeters 1996)	Open water, state W
MNDWI	Open water	(Xu 2006)	Primary open-water index (state W, A1)
AWEInsh, AWEIsh	Open water	(Feyisa et al. 2014)	Shadow-robust water extraction
WI2015	Open water	(Fisher et al. 2016)	Water classification check

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